

JAMES T. FERRARE

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<https://jamesferrare.github.io>

EDUCATION

Stanford University , Stanford, California <i>PhD (ongoing) in Biophysics</i>	2020 – Present
Tulane University , New Orleans, Louisiana <i>M.S. in Physics</i>	2016 – 2018
Tulane University , New Orleans, Louisiana <i>B.S. in Physics, Mathematics, and International Development; Magna cum laude</i>	2013 – 2017

RESEARCH OVERVIEW

Dissertation title: Non-equilibrium evolutionary dynamics in large adapting populations
Research description: I leverage tools from statistical physics and population genetics to develop simple theoretical models of microbial adaptation. I apply these models to understand how the horizontal exchange of genetic material influences long-term evolutionary outcomes in systems like the human gut microbiome.

PUBLICATIONS AND MANUSCRIPTS

The dynamics of horizontal gene transfer in rapidly adapting populations
James Ferrare, Daniel Wong, Daniel Fisher and Benjamin Good
Available upon request (Preprint expected 2026)

Evolution of evolvability in rapidly adapting populations
James Ferrare and Benjamin Good
Nature Ecology and Evolution (2024); doi: <https://doi.org/10.1038/s41559-024-02527-0>

Cholesterol in the cargo membrane amplifies tau inhibition of kinesin-1-based transport
Qiaochu Li, James Ferrare, Jonathan Silver, John Wilson, Luis Arteaga-Castaneda, Weihong Qiu, Michael Vershinin, Stephen King, Keir Neuman, and Jing Xu
Proceedings of the National Academy of Sciences (2023); doi: <https://doi.org/10.1073/pnas.2212507120>

Assembly of heteropolymers via a network of reaction coordinates
Davide Chiuchì, J. Ferrare, and Simone Pigolotti
Physical Review E (2019); doi: <https://doi.org/10.1103/PhysRevE.100.062502>

AWARDS AND FELLOWSHIPS

Dissertation Completion Fellowship, <i>Stanford University</i>	2026 – 2027
ARCS Pre-doctoral Fellowship, <i>Stanford University</i>	2024 – 2025
SMBE Young Investigator Travel Award, <i>Stanford University</i>	2023
Bio-X Travel Award, <i>Stanford University</i>	2022, 2023
Gates-Cambridge Scholarship Finalist, <i>Tulane University</i>	2018
J. William Fulbright Scholarship, <i>Tulane University</i>	2018
George C. Marshall Scholarship Finalist, <i>Tulane University</i>	2017
George J. Mitchell Scholarship Finalist, <i>Tulane University</i>	2017
Student Leaders in Service Award, <i>Tulane University</i>	2016
McKeever Civic Engagement Scholarship, <i>Tulane University</i>	2016
Stone Award for Best Undergraduate Paper on a Latin American Topic, <i>Tulane University</i>	2016
Newcomb Tulane Taylor Summer Travel Grant, <i>Tulane University</i>	2014, 2016
Conference Presentation Award, <i>Tulane University</i>	2016
TULASO Scholarship, <i>Tulane University</i>	2015
Stellar Tulane Academic Recognition Scholarship, <i>Tulane University</i>	2014

RESEARCH EXPERIENCE

Good Lab – Stanford University <i>Ph.D. Student</i> · Dissertation research	Stanford, California <i>January 2021 -</i>
Kavli Institute for Theoretical Physics <i>Student, Quantitative Biology School</i> · Sampled, sequenced, and characterized long-read soil phage genomes	Santa Barbara, California <i>Summer 2024</i>
Spakowitz Lab – Stanford University <i>Rotation student</i> · Developed Ising model of epigenetic interactions and chromatin phase dynamics	Stanford, California <i>Fall 2020</i>
Neuman Lab – National Institutes of Health <i>Post-baccalaureate fellow</i> · Modelled dynamics of fibrillar collagen proteolysis by matrix metalloproteinase 1 and 9 complexes	Bethesda, Maryland <i>2020 – 2022</i>
Tkačik Lab – Institute for Science and Technology Austria <i>Intern</i> · Investigated asymmetric partitioning of bacterial proteomes in stochastic antibiotic environments.	Klosterneuberg, Austria <i>2018 – 2019</i>
Pigolotti Lab – Okinawa Institute for Science and Technology <i>Intern</i> · Developed analytical error formula for DNA polymerization along arbitrary reaction coordinate	Okinawa, Japan <i>2018 – 2019</i>
Talbayev Lab – Tulane University <i>Undergraduate researcher</i> · Fabricated periodic nanostructures on InSb substrates using photolithography and electron beam deposition	New Orleans, Louisiana <i>2017 – 2018</i>

INTERNATIONAL DEVELOPMENT RESEARCH

Tulane University and Partners in Development <i>Principal Investigator</i> · Led two-year IRB-approved, longitudinal study of 750-patient chronic disease program in Haiti. · Spent 1000+ volunteer hours interviewing 80 patients and healthcare workers, constructing electronic health record system, analyzing collected data, piloting telemedicine program and publishing 140-page report of findings	Port au Prince, Haiti <i>2016 – 2018</i>
Tulane University and La Fundación Paraguay <i>Independent Study</i> · Led team of rural agricultural students to implement low-cost technology to build charcoal briquettes from agricultural waste · Wrote award-winning study on the impact of global energy poverty on multidimensional poverty metrics	Paraguay <i>2017 – 2018</i>

PROFESSIONAL EXPERIENCE

Scientific Consultant , StrandID · Developed proprietary physical binding model for pathogen testing device · Implemented in silico directed evolution algorithm to design custom genetic oligonucleotide probes	2026
Data Consultant , Withum Smith and Brown · Developed data-integration tools and interactive financial reports	2019-2020
Emergency Medical Technician , New Orleans Emergency Medical Services · Treated gun-shot wounds, heart attacks, overdoses, etc. on 12-hour volunteer ambulance shifts.	2015-2017

ORAL AND POSTER PRESENTATIONS

National Institute for Theory and Mathematics in Biology, <i>Chicago, Illinois, Poster presentation</i>	2025
Society for Molecular Biology and Evolution, <i>Puerto Vallarta, Mexico, Poster presentation</i>	2024
Society for Molecular Biology and Evolution, <i>Ferrara, Italy, Poster presentation</i>	2023
American Physical Society, <i>Las Vegas, Nevada, Oral presentation</i>	2023
Bay Area Population Genetics, <i>Berkeley, California, Poster presentation</i>	2023

Bay Area Population Genetics, <i>Stanford, California, Poster presentation</i>	2022
Population, Evolutionary and Quantitative Genetics, <i>Monterrey, California, Poster presentation</i>	2022
Tulane Physics Research Colloquium, <i>New Orleans, Louisiana, Poster presentation</i>	2017
Birmingham Southern Latin American Studies Symposium, <i>Birmingham, Alabama, Oral presentation</i>	2016

OUTREACH AND MENTORING

Mentor, Stanford University and The Tanzania Human Genetics Organization	Summer 2026 –
Tutor, George Washington High School	2025
Mentor, Nueva School Stanford Internship Program	Summer 2022, Spring 2023
Stanford Biophysics Executive Committee	2021
Tutor, Tulane University	2017-2018
Intern, New Orleans Child Advocacy Center	2017-2018
Intern, Covenant House New Orleans	2015